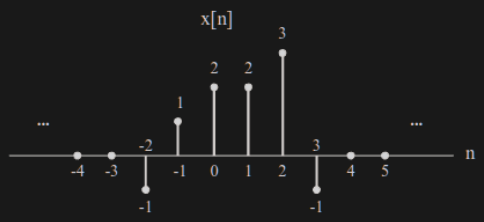
PROBLEMA 1.2

Dada la señal $x[n]$:



represente gráficamente las siguientes señales:

a.- $x[n+2]$

b.- $x[3-n]$

c.- $x[n^2-4]$

d.- $x[3n+2]$

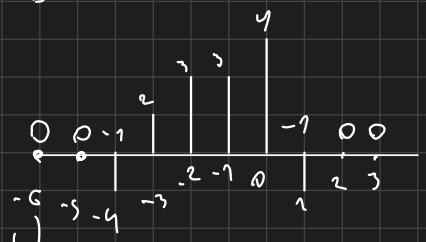
e.- $y[n]=\begin{cases} x\left[\dfrac{n}{2}\right] & para\ \ n\ \ par \\ 0 & para\ \ n\ \ impar \end{cases}$

f.- $x[-n+4]\cdot u[-n+2]$

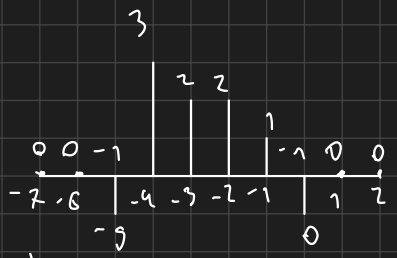
g.- $\left(\dfrac{1}{2}\right)^n \cdot x[n]$

h.- $x[n]\cdot\delta[n+2]+x[n-4]\cdot\delta[n-2]$

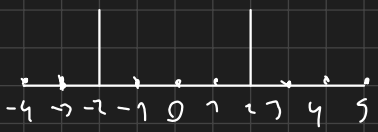
a)



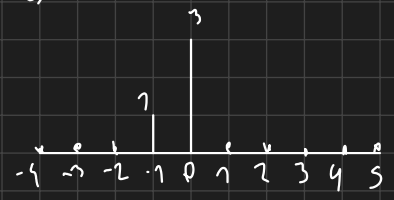
b)



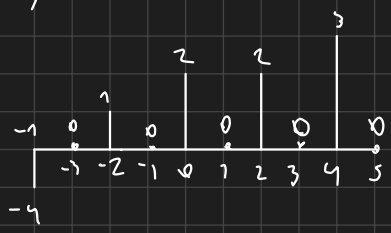
c)



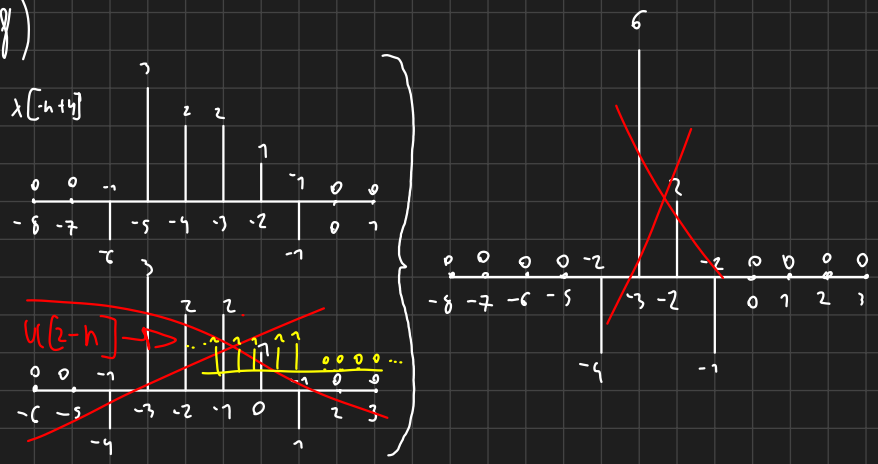
d)



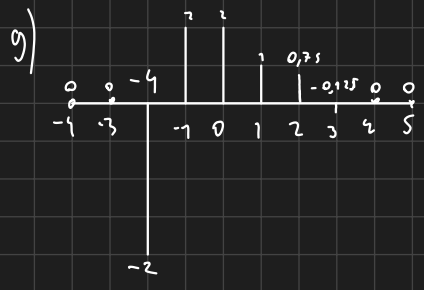
e)



g)



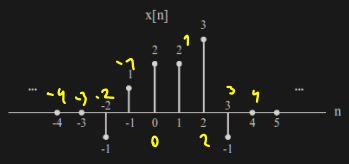
h)



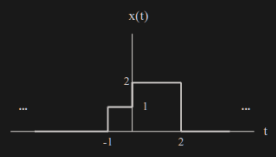
PROBLEMA 1.3

Descomponga las siguientes señales en sus componentes par e impar, y compruebe que ambas cumplen sus propiedades de simetría.

a)



b)



* Par, $x[n] = x[-n]$

Impar, $x[n] = -x[-n]$

$$X_e = \frac{x[n] + x[-n]}{2}$$

$$X_o = \frac{x[n] - x[-n]}{2}$$



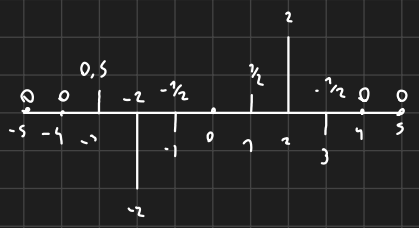
a) $x_e[-4] = 0$ $\quad x_o[-4] = 0$

$x_e[-3] = \frac{1}{2}$ $\quad x_o[-3] = \frac{1}{2}$

$x_e[-2] = 1$ $\quad x_o[-2] = -2$

$x_e[-1] = \frac{3}{2}$ $\quad x_o[-1] = \frac{-1}{2}$

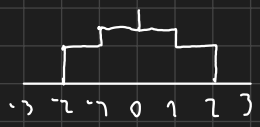
$x_e[0] = 1$ $\quad x_o[0] = 1$

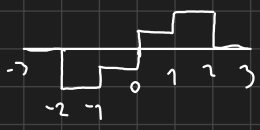


b) $x_e(-2) = 1$ $\quad x_o(-2) = -1$

$x_e(-1) = 3/2$ $\quad x_o(-1) = -\frac{1}{2}$

$x_e(0) = 2$ $\quad x_o(0) = 0$





PROBLEMA 1.4

Determine si las siguientes señales son periódicas o no. En caso afirmativo, obtenga su periodo fundamental.

a.- $x_1(t) = 5 \cdot sen\left(6t + \frac{\pi}{8}\right)$

b.- $x_2(t) = sen(3\pi t) \cdot u(t)$

c.- $x_3(t) = 4 \cdot sen\left(\frac{\pi}{6}t + \frac{\pi}{8}\right) + 2 \cdot \cos\left(\frac{\pi}{4}t + \frac{\pi}{3}\right)$

a) $w = \frac{2\pi}{T} \;//\; T = \frac{2\pi}{w} \;//\; t = \frac{2\pi}{6} = \frac{\pi}{3}$

b) por ser 0 desde $(-\infty, 0)$, no puede ser periódica ya que no lo es en $(-\infty, 0)$

c) Sera periódica, cogemos t_{mayor}, periodo = MCM

$$\left.\begin{array}{l} T\ \frac{2\pi}{\frac{\pi}{6}} = 12 \\ T\ \frac{2\pi}{\frac{\pi}{4}} = 8 \end{array}\right\} \quad 8 \cdot 3 = 12 \cdot 2 = \boxed{24}$$

PROBLEMA 1.5

Determine si las siguientes secuencias son periódicas o no. En caso afirmativo, obtenga su periodo fundamental.

a.- $x_1[n]=\cos\left(\frac{4\pi n}{5}+1\right)$

b.- $x_2[n]=sen\left(\frac{n}{3}\right)\cdot sen\left(\frac{\pi n}{8}\right)$

c.- $x_3[n]=e^{j\left(\frac{n}{3}-\frac{\pi}{4}\right)}$

$\nu=\frac{2\pi}{T}\ //\ T=\frac{2\pi}{\omega}\ //\ T=\frac{2\pi}{\frac{4\pi}{5}}$